

Analyzing use of masculine generics by LLMs in French

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1. Introduction

- Masculine generics (MG) in gender-marked languages (e.g., French, German, Dutch): use of the masculine form as a default/neutral form to refer to (a) mixed group of men/women or (b) people whose gender is unknown. Examples in French:

(a) « Les **étudiants** sont partis. » (**Students** [masc.] left.)

(b) « Un **athlète** doit s’entraîner régulièrement pour progresser. » (An **athlete** [masc.] needs to train regularly to progress.)
- Psycholinguistics studies show that **MG induce gender bias** and **amplify male-centric mental representations**^[1–3]
- Gender bias widely studied in instruct-based LLMs^[4], but never with generic instructions or in unconstrained contexts

2. Methodology

Focus on French, but applicable to any language with MG given human noun and instruction datasets

- a. Create a French human noun (HN) dataset from available French lexical resources to detect occurrences of MG and evaluate the ratio of MG to HN uses
- Filter with custom ML binary HN classification pipeline
- Golden

Filtered

Not used for classification

NHUMA

Wikidata

Wiktionary

Demonette

TLFI (Recursive)

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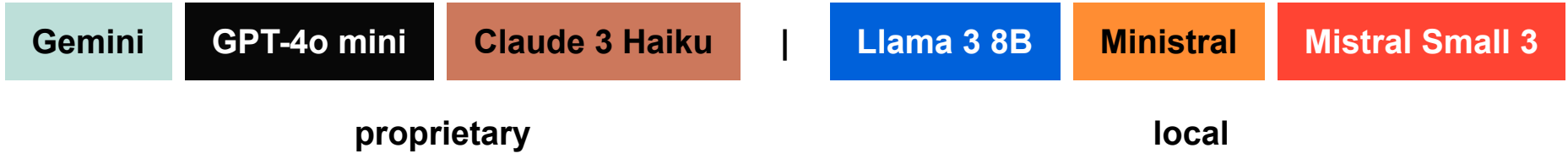
TLFI
- b. Analyze MG use in 6 LLMs’ outputs to generic instructions
- Use human/AI-written generic instruction datasets and remove specific contexts with spaCy^[5] (dependency parsing and NER)

Pourquoi Paris est si populaire ?
(Why is Paris so popular?)

Comment fixer une télé au mur ?
(How to mount a TV to the wall?)

Qui est Albert Camus ?
(Who is Albert Camus?)

- Sample 10,000 instructions and send them to LLMs



- Like instructions, filter responses to remove specific contexts
- Validate HNs in outputs using GPT-4o mini, JSON-constrained
- Compute score for each text; as well as mean (average bias per text) and overall (average bias per LLM) scores

$$MScore_i = \frac{mg_count_i}{hn_count_i}$$

Overall

$$\frac{total_mg}{total_hn}$$

Mean

$$\frac{1}{n} \sum_{i=1}^n MScore_i$$

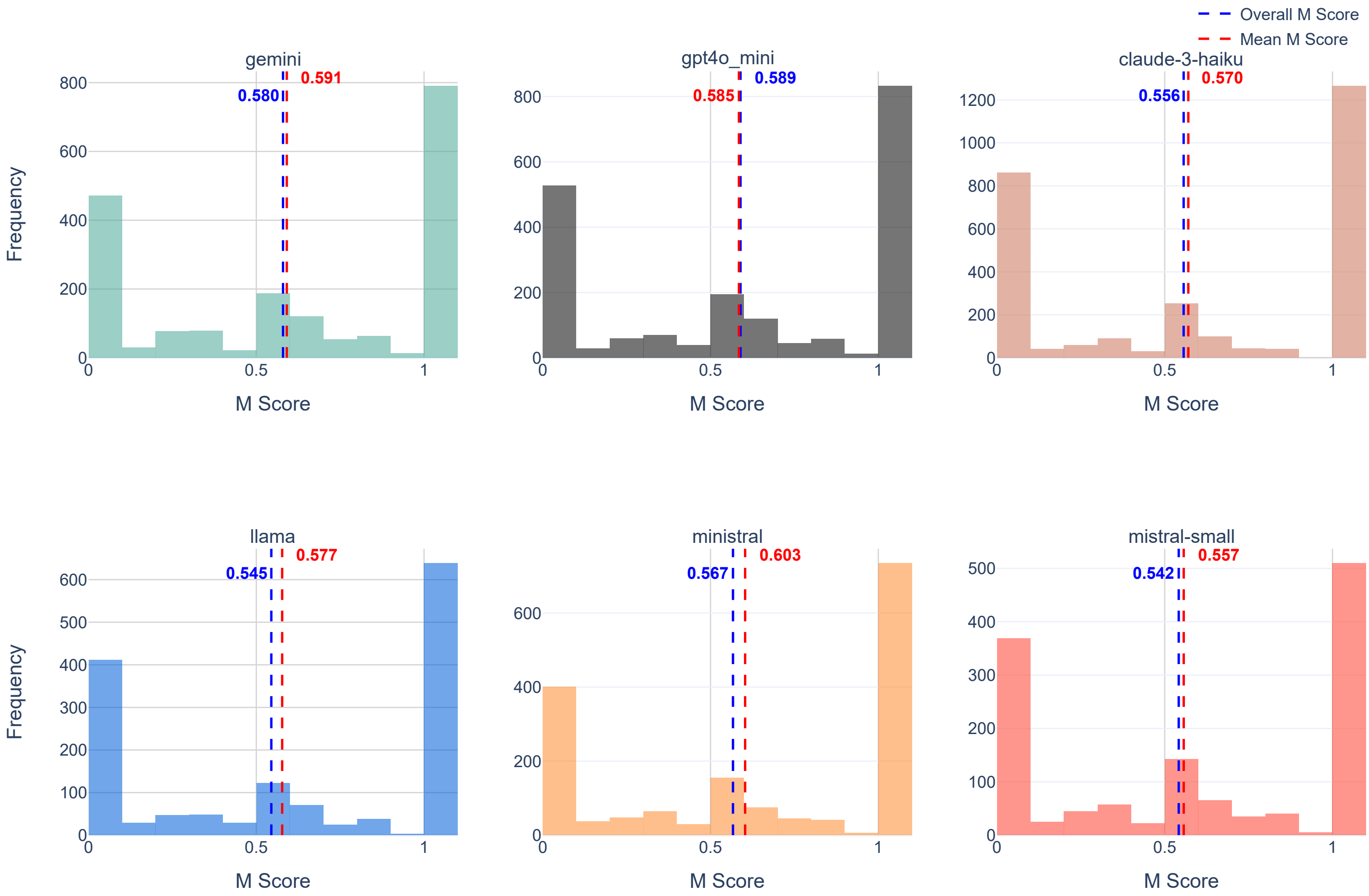
References

[1] Braun, F. et al. (2005). Cognitive Effects of Masculine Generics in German. *Communications*, 30(1).
[2] Gygax, P. et al. (2012). The Masculine Form and Its Competing Interpretations in French. *Journal of Cognitive Psychology*, 24(4).
[3] Rothermund, P., & Strack, F. (2024). Reminding May Not Be Enough. *Journal of Language and Social Psychology*, 43(4).
[4] Gallegos, I. O. et al. (2024). Bias and Fairness in Large Language Models. *arXiv*.
[5] Honnibal, M. et al. (2023). Explosion/spaCy.

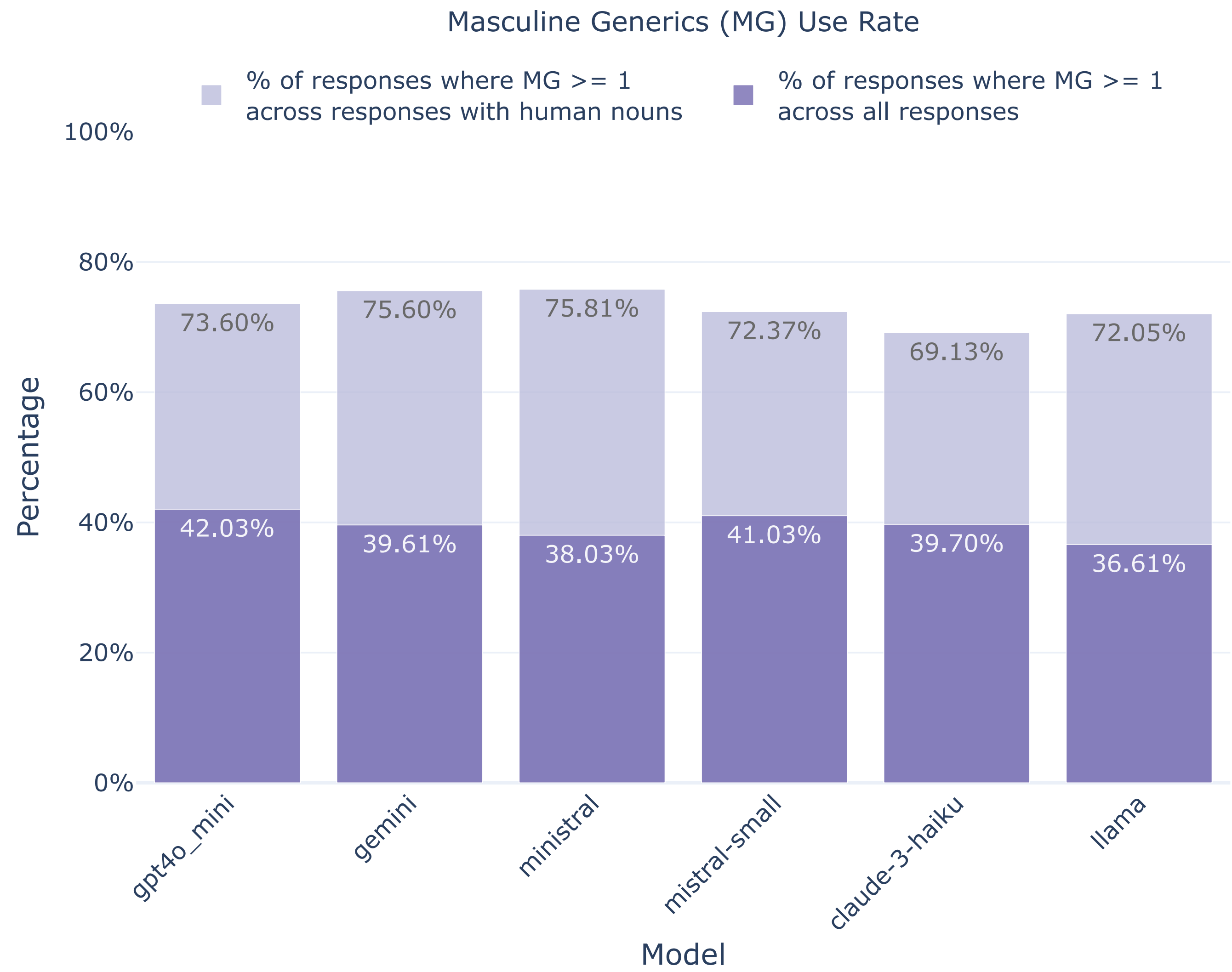
3. Results and Findings

- LLMs use MG in $\approx 39.5\%$ of all their responses on average ($\approx 73.1\%$ of responses with HNs)
- GPT-4o mini and Ministral generally the most biased models
- Llama 3 8B, Claude 3 Haiku and Mistral Small 3 generally the least biased models
- LLMs reluctant to using gender-fair language (GFL) spontaneously, Llama 3 8B being the model with highest GFL use (see preprint for details)

MScore



MG Use Rate



4. Takeaways

- LLMs largely exhibit MG bias when generating responses to generic, contextually unconstrained instructions
- Fairness in language should be attentively considered when training LLMs in heavily gender-marked languages

Read our full preprint for more details:

Doyen, E. & Todirascu, A. (2025). *Man Made Language Models? Evaluating LLMs’ Perpetuation of Masculine Generics Bias*. *arXiv*: 2502.10577.

